

TRAVEL & EXPENSE MANAGEMENT

Process Analytics, Data Mining & Process Redesign

Intermediate Report Phase 2: In-depth Analysis

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Reported by Group 6:

Executive Summary

This report presents an in-depth analysis of UHasselt's travel and expense reimbursement system, covering the academic years 2017 and 2018. The analysis was conducted on five administrative datasets comprising 270,211 recorded events across 33,049 unique process cases, using the CRISP-DM framework implemented in KNIME, Disco and visualised through a Power BI dashboard.

The university operates a multi-stage approval process through which academic and administrative staff submit travel authorisations and expense claims for professional trips. The process faces challenges around compliance verification, non-standard case behaviour, and inefficient processing of permits linked to multiple expense claims despite being a well-established administrative system.

Three research questions were investigated: the extent to which expense claims are correctly preceded by approved travel authorisations, whether anomalous or risky behaviour can be detected in the data, and how the end-to-end process can be redesigned to improve throughput efficiency. The analysis reveals that the system's primary financial control, the permit authorisation requirement does not reliably prevent unauthorised expenditure. A significant proportion of claims were processed and paid without valid authorisation, budget limits are routinely exceeded, and submission deadlines are frequently missed. These issues are not isolated incidents but reflect structural weaknesses in how the approval process is designed.

A process redesign is proposed that consolidates independent claim approvals into a single coordinated review per trip, directly addressing the identified compliance and efficiency gaps. Four recommendations were made; embedding a pre-payment permit validity check, enforcing submission deadlines through system controls, implementing the bundled approval process, and introducing independent oversight when approval roles conflict.

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1. Business Understanding

1.1 Background and Organisational Context

Hasselt University operates a structured travel and expense reimbursement system for its academic and administrative staff. Employees frequently travel for professional purposes, including attendance at international and national conferences, visits to partner universities, collaboration with researchers, and participation in project meetings. Because these trips generate substantial costs, the university has established formal procedures for approving travel in advance and reimbursing expenses afterwards.

The data used in this project originates from an anonymized administrative dataset covering the years 2017 and 2018. It is important to note that the year 2017 represents a trial phase of the system, with only two departments participating. Consequently, the processes observed in 2017 may differ structurally from those in 2018, which includes all university departments. Furthermore, the underlying processes have been revised multiple times since the initial trial, making direct year-on-year comparisons potentially misleading without accounting for these changes.

1.2 Process Overview

There are two overarching process flows can be identified within the university's travel and expense management system:

1.2.1 Travel Authorization Process (For International Travel Only)

International travel requires prior approval before a trip can take place. The employee submits a travel permit, which must be approved before any travel occurs. No payment is involved in this process. The travel permit includes estimated start and end dates, which are provided by the employee at submission time and represent planned rather than confirmed dates. Actual travel dates are not recorded in the data.

After a travel authorization is approved:

- Employees may submit prepaid travel expense requests before the trip begins.
- Multiple prepaid expense requests can be submitted independently per authorization.
- Following the trip, employees receive reminders to submit an international expense claim.
- Multiple international expense claims may be submitted for a single authorization.

1.2.2 Expense Claim Approval Flow (All Request Types)

All expense-related claims follow a broadly consistent approval process. The general sequence follow these steps:

Step	Description
1. Submission	Employee submits the request (domestic claim, international claim, prepaid expense, or payment request).
2. Travel Administration Review	The request is reviewed by the travel administration team. Approval forwards the request; rejection triggers a resubmission or final rejection.
3. Budget Holder Approval	The approved request is forwarded to the budget holder for financial authorization.
4. Supervisor Approval	The request is reviewed by the supervisor. If the budget holder and supervisor are the same person, only one approval step is needed.
5. Director Approval (Conditional)	In specific cases, an additional approval by the director is required.

Step	Description
6. Payment	Once all approvals are obtained, the payment is requested and processed.
7. Rejection Handling	At any step, a request may be rejected. The employee may resubmit a revised request, or the request may be definitively closed.

1.3 Request Types

The dataset encompasses four distinct request types, each with slightly different characteristics and contexts:

Request Type	Description
National Expense Claims	Submitted by employees after domestic travel. No prior travel permit required. Claims can be submitted up to two months after the trip.
International Expense Claims	Submitted after international travel. Must be linked to an approved travel authorization. Claims can cover accommodation, meals, and other on-site expenses.
Prepaid Travel Expenses	Submitted before or during an international trip (e.g., flights, conference registration fees). Linked to an approved travel authorization.
Payment Requests	Requests for reimbursement of work-related purchases not directly related to travel (e.g., office equipment, materials).

1.4 Research Questions

This project addresses three interconnected research questions that span process analytics, data mining, and process redesign:

RQ 1: Process Analytics (Compliance)

- To what extent are expense claims correctly preceded by an approved travel authorization?
- Are there cases where expense claims exist without a corresponding authorization?

RQ 2: Data Mining (Anomaly Detection)

- Can anomalies or outliers be detected that may indicate non-standard or risky behavior?

RQ 3: Process Redesign (Throughput Optimization)

- How can the end-to-end throughput time be improved when multiple expense claims are linked to a single travel authorization?
- How would this redesigned process be represented in BPMN?

1.5 Business Objectives

This project seeks to translate the research questions into actionable analytical objectives, specifically aiming to:

- Evaluate the alignment between the travel authorization process and the submission of expense claims, identifying any structural gaps or non-compliance issues.
- Quantify the proportion of international expense claims lacking a valid prior travel authorization, and analyze the characteristics of these cases.

- Identify potential anomalies within the data through process mining and statistical analysis, determining if these patterns suggest procedural discrepancies.
- Assess the efficiency of the current approval process, particularly for travel permits associated with multiple expense claims, and propose a refined process model for improvement.

1.6 Success Criteria

The following success criteria outline the standards for a satisfactory outcome for each of the set business objectives:

Objective	Success Criterion
Compliance Assessment	A quantified compliance rate for travel authorizations preceding international expense claims, with case-level identification of non-compliant records.
Anomaly Detection	A documented set of anomalies with supporting statistical or process evidence, categorized by type and potential risk level.
Process Redesign	A BPMN model that illustrates a redesigned process with measurable expected improvements in throughput time, supported by a logical rationale.
Exploratory Analysis	A structured EDA covering all five datasets, identifying data quality issues, distributions, and key patterns relevant to the research questions.

1.7 Key Stakeholders

Understanding the key stakeholders involved in the travel and expense management process is essential for identifying decision-making authority, approval dependencies and the impact of the process inefficiencies on different user groups.

Stakeholder	Role & Interests
Employee (Academic/Administrative Staff)	Initiates travel permits and expense claims. Primary interest includes fast reimbursement, minimal administrative burden, and clear guidance on process rules.
Travel Administration	First reviewer of all requests. Responsible for completeness checks and policy compliance. Interest includes accurate submissions and reduced rework from incomplete claims.
Budget Holder	Responsible for financial authorization. Approves or rejects based on budget availability and interested in budget adherence and cost transparency.
Supervisor/Manager	Approves international travel permits and expense claims. Interested in oversight of staff travel activity.
Director	Provides additional approval in specific escalation cases and interested in governance, compliance with high-value or exceptional cases.
Finance/Payment Processing	Executes payments once all the approvals are in place. Interested in complete and valid documentation before payment.
University Management	Oversees university-wide travel expenditure and policy compliance. Interested in cost control, process efficiency and risk management.
IT/System	Operates and maintains the administrative system. Some processes are executed automatically by the system (e.g., reminders). Interested in system reliability and data integrity.

1.8 Business and Reimbursement Rules

The following rules govern the travel and expense reimbursement process at Hasselt University. These rules define compliance boundaries and form the normative baseline against which the data will be evaluated.

1.8.1 Authorization Rules

- International travel requires a travel permit to be submitted and approved by the employee's manager before the trip takes place. Travel undertaken without prior approval is non-compliant.
- Domestic travel does not require a travel permit. Employees may travel and subsequently submit a national expense claim without prior authorization.
- The travel permit must include estimated start and end dates for the trip. These represent planned dates; actual dates are not recorded.
- Once a permit is approved, the employee may submit requests for prepaid travel expenses (e.g., flights, registration fees) before the trip begins.

1.8.2 Expense Claim Submission Rules

- Expense claims may be submitted as soon as costs are incurred (e.g., for pre-booked flights or conference registration) or within two months after the trip ends.
- International expense claims must be linked to an approved travel authorization. A claim without a corresponding approved permit is non-compliant.
- Multiple expense claims may be submitted per travel authorization (e.g., one for flights, one for accommodation). Each is processed independently through the approval workflow.
- National expense claims are processed independently and do not require linkage to any travel permit.
- Payment requests (non-travel purchases) follow the same approval workflow but are not linked to any travel authorization.

1.8.3 Approval Flow Rules

- All expense-related requests must pass through travel administration review before proceeding to financial approval.

- If the budget holder and supervisor are the same individual, only one approval step is required; the duplicate step is collapsed.
- In specific cases (criteria not fully specified in the data description), an additional director approval is mandatory before payment can be processed.
- Rejection at any approval stage results in either: (a) the employee revising and resubmitting the request, or (b) the request being definitively rejected and closed.
- Payment may only be initiated after all required approvals are obtained. No partial payments are described in the process.

1.8.4 Post-Trip Compliance Rules

- After a travel authorization is approved and the trip has taken place, employees receive system-generated reminders to submit a travel expense report.
- Failure to submit an expense claim after an approved trip (and after reminders) may indicate a process gap or an abandoned claim; an area of interest for compliance analysis.
- Monetary amounts in the data are anonymized and not exact. However, for sufficiently large subsets, totals and averages remain representative and can be compared to approved budgets.

1.9 Business Problem and Challenges

Based on the description of the organization and the data structure, the following business problems can be anticipated. They represent the practical challenges the university faces and motivates the analytical objectives of this project.

BP 1: Data Quality and System Inconsistencies (2017 vs. 2018)

- The data from 2017 reflects a trial phase with only two departments, and the underlying process rules changed multiple times after the trial. This introduces structural inconsistencies that complicate cross-year comparisons.
- **Business impact**: Analytical results that do not account for this structural break may lead to misleading conclusions about trends or compliance rates.

- **Analytical focus**: Clearly separating and documenting the 2017 and 2018 datasets during analysis; flagging where process rule changes may explain observed differences.
- **RQ:** How did structural inconsistencies (limited departments, process changes) in the 2017 trial data affect observed trends and compliance versus the 2018 dataset, and how should future analyses adjust for this structural break?

BP 2: Independent Audit Committee

- An independent, separate committee is responsible for verifying that submitted permits and authorizations meet the required credibility standards of the business process.Based on the highlighted text:
- **RQ:** How does the organizational structure and operational independence of the verification committee affect its ability to enforce "required credibility standards" for permits and authorizations?

BP 3: Rework and Delays

- This shows a lack of understanding from the employee regarding the business process requirements, leading to a lot of rework and delays.
- **RQ:** How does the employee's lack of understanding of business process requirements correlate with the frequency of rework and project delays in the workflow?

BP 4: Conflicting Roles

- The business process shows the supervisor acting sometimes as the budget holder, which may lead to contradicting decisions and accountability issues.
- **RQ:** What are the potential risks and implications of the supervisor occasionally acting as the budget holder in the current business process, particularly regarding decision-making consistency and accountability?

2. Data Understanding

This section presents the findings from the exploratory data analysis conducted on all five provided datasets. Each dataset is profiled in terms of structure, data quality, and key distributional patterns, followed by an integrated view of cross-dataset relationships.

Exploratory Data Analysis

One of the following analyses was conducted to ensure a comprehensive evaluation and accurate analysis of each dataset.

- Record counts and column inventory (names, types, nullability).
- Missing Data Analysis: Examination of absolute counts and proportions for each column with missing values.
- Distribution Analysis: Histograms for numerical variables and frequency tables for categorical variables.
- Temporal Analysis: Evaluation of submission and approval volumes over time, broken down by month or quarter.
- Cross-Dataset Linkage Analysis: Assessment of match rates between travel permits and corresponding expense claims.
- Duplicate Detection: Identification of cases with identical key field values.
- Outlier Detection: Identification of extreme values in financial amounts and processing times.

2.1 Travel Permits

The Travel Permits dataset serves as the central authorization record for all international travel at Hasselt University. It is the largest and most structurally complex of the five datasets, originally stored in a wide format that required reshaping into a standard event log before analysis could proceed.

2.1.1 Dataset Snapshot

Attribute	Value
Total rows (events)	86,581
Total unique cases (permits)	7,065
Time period	2016-10-05 to 2021-09-01
Original column count	175 (wide format)
Columns after cleaning	9 (event log format)
Unique activities	51
Missing values	None
Duplicate rows	None

The 175-column wide format reflects an indexed structure where each permit stores up to 17 linked declaration references as separate numbered columns. The majority of these indexed columns are empty for most permits only permits with multiple linked claims populate the higher-indexed columns. This is a structural design choice, not a data quality issue, but it required reshaping to a long-format event log before any activity-level analysis could be performed.

2.1.2 Activity Analysis

The 51 unique activities represent the full lifecycle of a travel authorization, from initial employee submission through to final payment or rejection. The six most frequent activities are shown below.

Activity	Count
Declaration SUBMITTED by EMPLOYEE	7,574
Payment Handled	7,544
Request Payment	7,541

Permit SUBMITTED by EMPLOYEE	7,331
End trip	7,065
Start trip	7,065

Every permit has a recorded Start trip and End trip (both 7,065) confirming that trip execution is consistently logged for all cases. The near-identical counts for Request Payment (7,541) and Payment Handled (7,544) indicate that almost all payment requests result in a completed payment with minimal dropout at the final stage. The most analytically significant finding is that Declaration SUBMITTED by EMPLOYEE (7,574) outnumbers Permit SUBMITTED (7,331) by 243 events. This means some employees submitted multiple declarations per permit, directly evidencing the multi-claim behaviour that is the core subject of RQ3. Payment Handled (7,544) also exceeds Permit SUBMITTED (7,331), further confirming that multiple payment events are generated per permit in a subset of cases. Rejection activities account for 1,912 events (2.2% of all 86,581 events). While this appears small as a percentage, it means approximately 1 in 4 permits experienced at least one rejection during its lifecycle, confirming that rework and resubmission are structurally embedded in the process rather than rare exceptions.

2.1.3 Processing Time

Processing time is measured from the first recorded event to the last recorded event per permit case.

Metric	**Value (≈days)**
Mean throughput	87
Minimum throughput	0
Maximum throughput	1,190
Median throughput	71
Std deviation	66 days

The mean throughput of 87 days is 22.5% higher than the median of 71 days, which indicates that a subset of extreme cases with very long processing times are pulling the average upward while the typical permit resolves considerably faster. The gap is meaningful but not as severe as initially estimated; most permits complete within approximately 10 weeks, but the tail end of the distribution extends dramatically to a maximum of 1,190 days. A standard deviation of 66 days against a mean of 87 days confirms that the high variability permit processing time is far from consistent across cases. A permit taking 1,190 days represents over three years of processing time for a single travel authorisation, a finding that cannot be explained by normal process delays and is investigated as a critical anomaly in section…

2.1.4 Key Data Quality Findings

- **350 permits (4.8% of all permits) never reached final approval.** Of the permits that did complete the process, 6,300 received final supervisor approval and 681 were escalated to director level; confirming two valid approval routes exist within the standard process. The 350 incomplete permits represent a governance gap where employees may have travelled without formally completing the travel authorisation required by the university. While 4.8% appears small, it represents 350 cases where expenditure may have been incurred without verified organisational approval.
- **Travel permit number 424** has **438 unique linked expense claims**. Having some declarations with about 17 different activities and a mean of 7.64 claims per multi-claim permit, making the permit more complex than average. This single permit is the most extreme throughput case in the dataset and the primary driver of the multi-claim inefficiency that will be investigated in RQ3.
- **Wide-format indexed columns create structural sparsity.** The original 175-column structure in this table stores up to 17 linked declaration references per permit as numbered columns. For single-claim permits, the majority of the columns were empty. This is a business implication as the multi-claim permit behaviour needs the table to be reshaped first in order to deflate the wide format of the columns from 175 to 9 meaningful columns.

2.1.5 Initial Insights

This table is the foundation for all the three research questions and the 350 permits without the final approval inform RQ1 compliance analysis. The maximum throughput of 1,190 days and permit number 424 with 438 linked claims are the primary anomalous cases for RQ2. The confirmed throughput difference between multi-claim & single-claim permits are more interesting for the RQ3 process redesign.

2.2 International Expense Claims

The International Expense Claims dataset records all post-trip reimbursement submissions for international travel at Hasselt University. It is the second largest dataset by event count and the only table with a mandatory foreign key link to the Travel Permits dataset, making it central to the RQ1 compliance analysis.

2.2.1 Dataset Snapshot

Attribute	**Value**
Total rows (events)	72,151
Total unique cases (permits)	6,449
Time period	2016-05-19 to 2020-05-10
Columns	25
Unique activities	34
UNKNOWN permit links	218
Most linked permit	Travel permit 424

2.2.2 Activity Analysis

34 unique activity types reflect the full declaration lifecycle from initial employee submission through administration, budget owner, supervisor, and director approval stages to final payment. The ten most frequent activities are shown below.

Activity	Count
Declaration SUBMITTED by EMPLOYEE	8,099
End trip	6,449
Start trip	6,449
Permit SUBMITTED by EMPLOYEE	6,255
Payment Handled	6,187
Request Payment	6,183

The consistent counts for Start trip and End trip (6,449) confirm that every claim case has a recorded trip execution equal to the total case count. This means trip logging is complete with no gaps across all 6,449 cases. Declaration SUBMITTED by EMPLOYEE (8,099) exceeds the total case count of 6,449 by 1,650 events; confirming that a subset of employees submitted multiple declarations per trip, directly evidencing the multi-claim behaviour investigated in RQ3.

2.2.3 Processing Time

Processing time is measured from the first recorded event to the last recorded event per permit case.

Metric	Value (≈days)
Mean throughput	86
Median throughput	66
Minimum throughput	6
Maximum throughput	742

The mean throughput of 86.1 days is 30.5% higher than the median of 66 days, indicating that a subset of extreme cases is pulling the average upward while the typical claim resolves in approximately nine weeks. The minimum of 6 days contrasts sharply with the maximum of 742 days, a range of over two years

between the fastest and slowest cases. The close alignment between the mean throughput here (86 days) and the Travel Permits mean (87 days) is expected international claims are processed in conjunction with their linked permits and therefore follow a similar timeline.

2.2.4 Key Data Quality Findings

- **218 cases (3.4% of all claims) have UNKNOWN as permit link.** These claims cannot be traced to an approved travel authorisation. It means 218 international expense claims were reimbursed or are pending reimbursement without a verifiable authorisation record; this is a direct compliance risk and serves as the primary non-compliant case group for RQ1.
- **Travel permit 424 with 3,377 linked claim events.** This one permit accounts for 4.7% of all events in this dataset. The extreme concentration of claims on a single permit distorts aggregate processing time statistics and is the most extreme multi-claim case for RQ3 throughput analysis.
- **4,192 rejection events (5.8% of all 72,151 events).** The rejection activities visible in the Row Filter node output include Declaration REJECTED by PRE_APPROVER, Declaration REJECTED by EMPLOYEE, Declaration REJECTED by MISSING, Declaration REJECTED by SUPERVISOR and Permit REJECTED by MISSING; confirming that rejections occur at multiple stages of the approval chain. The implication for the business is that the rework is structurally embedded in the international claims process, contributing directly to the extended mean throughput time.

2.2.5 Initial Insights

In the international expense claims table, the dataset is mostly analytically connected as its findings directly address the three research questions. The 218 UNKNOWN permit links serve as the primary compliance gap for RQ1. The 4,192 rejected events and the extreme throughput range of 6 to 742 days provide the strongest anomaly signals for RQ2 while the 3,377 events on a single permit confirms the multi-claim processing inefficiency in RQ3.

2.3 National Expense Claims

The National Expense Claims The National Expense Claims dataset records all domestic expense reimbursement submissions at UHasselt. Unlike international claims, domestic submissions do not require a prior travel permit; employees submit directly for reimbursement without prior authorisation. This makes it the simplest and fastest process in the dataset, and the only table with no mandatory foreign key linkage to any other table.

2.3.1 Dataset Snapshot

Attribute	**Value**
Total rows (events)	56,437
Total unique cases (permits)	10,500
Time period	2017-01-09 to 2019-06-17
Columns	12
Unique activities	17
Rejection counts	2,846

2.3.2 Activity Analysis

There are 17 unique activities that cover the full domestic claim lifecycle from employee submission through administration, budget owner, pre-approver, and supervisor approval to final payment. The top 5 activities are shown below;

Activity	**Count**
Declaration SUBMITTED by EMPLOYEE	11,531
Declaration FINAL_APPROVED by SUPERVISOR	10,131
Payment Handled	10,044
Request Payment	10,040
Declaration APPROVED by ADMINISTRATION	8,202

Declaration SUBMITTED by EMPLOYEE (11,531) exceeds the total case count of 10,500 by 1,031 events confirming that some employees could have saved and resubmitted declarations more than once before final submission. The near-identical counts for Payment Handled (10,044) and Request Payment (10,040) indicate that almost all payment requests result in completed payment with minimal dropout at the final stage, consistent with the pattern observed in Travel Permits and International Claims.

2.3.3 Processing Time

Processing time is measured from the first recorded event to the last recorded event per permit case.

Metric	**Value (≈days)**
Mean throughput	11
Median throughput	7
Minimum throughput	0
Maximum throughput	469
Standard deviation	17

The mean throughput of 11 days is significantly faster than both Travel Permits (87 days) and International Claims (86 days); confirming that the absence of an authorisation stage significantly reduces cycle time for domestic claims. The median of 7 days means the typical domestic claim resolves within one working week. However, the mean of 11 days is 60% higher than the median of 7 days. Although the maximum of 469 days is quite extreme for a process that typically resolves in under two weeks. A standard deviation of 17 days, which is larger than the mean itself, confirms that a small number of outlier cases are severely distorting the average. timeline.

2.3.4 Key Data Quality Findings

- **2,846 rejection events (5.0% of all 56,437 events) across six rejection activity types.** Several rejections occur at administration, budget owner, supervisor, pre-approver, missing, and employee levels. This implicates the business as the rework is present at every stage of the domestic claim approval chain, not just at one bottleneck step which suggests systemic submission quality issues rather than a single approval stage being too strict.
- **'Declaration REJECTED by MISSING' accounts for 91 events across the dataset.** MISSING is a process step that was performed by a resource not registered in the system. These events cannot be audited or attributed to a responsible party, representing a business implication in the domestic claims process.
- **Maximum throughput of 469 days for a domestic claim.** Given that the typical claim resolves in 7 days, a case taking 469 days (over 15 months) cannot be explained by normal processing delays. This is one of the strongest single anomaly candidates in the National Claims dataset.

2.3.5 Initial Insights

The 7-day median throughput and simple 17-activity structure represents what an efficient approval process looks like when permit authorisation is not required. The 2,846 rejection events and the 469-day maximum throughput are the primary anomaly signals for RQ2. The speed difference between domestic claims (11 days mean) and international claims (86 days mean) quantifies the cost of the sequential permit-linked approval chain directly informing the parallel processing redesign proposed for RQ3.

2.4 Travel Expense

The Travel Expenses dataset records prepaid costs submitted before international trips; typically flights, conference registration fees, and accommodation booked in advance. Unlike post-trip expense claims, prepaid requests must be submitted and approved before the trip takes place and must be linked to an approved travel permit.

2.4.1 Dataset Snapshot

Attribute	Value
Total rows (events)	18,246
Total unique cases (permits)	2,099
Time period	2017-01-09 to 2019-02-21
Columns	24
Unique activities	29

2.4.2 Activity Analysis

The 29 unique activities cover both the prepaid payment request lifecycle and the permit authorisation steps that precede it. The top activities are shown below;

Activity	Count
Request For Payment SUBMITTED by EMPLOYEE	2,279
Payment Handled	1,990
Request Payment	1,990
Request For Payment FINAL_APPROVED by SUPERVISOR	1,958
Permit SUBMITTED by EMPLOYEE	1,952
Request For Payment APPROVED by ADMINISTRATION	1,726
Permit APPROVED by ADMINISTRATION	1,603
Permit FINAL_APPROVED by SUPERVISOR	1,478

Both the Request For Payment and Permit activities confirm that Travel Expenses captures the full prepaid cycle from permit submission through to payment in a single event log. This distinguishes it from the National Claims table, which contains only declaration activities. Request For Payment SUBMITTED by

EMPLOYEE (2,279) exceeds the total case count of 2,099 by 180 events, confirming that some employees submitted multiple prepaid requests per case. The identical counts for Payment Handled (1,990) and Request Payment (1,990) indicate no dropout at the final payment stage.

2.4.3 Processing Time

Processing time is measured from the first recorded event to the last recorded event per permit case.

Metric	**Value (≈days)**
Mean throughput	37
Median throughput	24
Minimum throughput	0
Maximum throughput	324

The mean throughput of 37 days is faster than both Travel Permits (87 days) and International Claims (86 days) but considerably slower than National Claims with (11.2 days). The median of 24 days means the typical prepaid request resolves within approximately three and a half weeks, which is reasonable given that prepayment must be processed before the trip departure date. The maximum of 324 days for a prepaid expense that should be resolved before a trip takes place cannot be explained by normal processing delays and is an anomaly candidate for RQ2.

2.4.4 Key Data Quality Findings

- **887 events belonging to UNKNOWN-linked cases.** Travel Expenses requires a mandatory link to an approved permit, yet a subset of cases show UNKNOWN in the permit number field across all permit-related metadata columns. These cases cannot be verified as legitimately authorised because prepaid expenses were processed without a traceable permit link, representing a direct compliance risk for RQ1.
- **749 rejection events (4.1% of all 18,246 events) across multiple rejection activity types.** Rejections occur at MISSING, PRE_APPROVER, and EMPLOYEE levels as visible in the Row Filter output. This is an

implication for the business as the presence of MISSING role rejections steps performed by unidentified actors, means a subset of rejection decisions cannot be attributed to a responsible approver, creating an audit gap.

- **Maximum throughput of 324 days for a prepaid expense.** Prepaid requests by definition must be approved before the trip, a case taking 324 days represents a fundamental process failure where either the prepayment was requested far in advance or the case became stuck in the approval chain indefinitely. This case is the strongest single throughput anomaly in the Travel Expenses dataset.

2.4.5 Initial Insights

Travel Expenses dataset is critical for compliance because every prepaid expense is required by university policy to be linked to an approved travel permit. Any case showing UNKNOWN in the permit field means this rule was not followed, making these cases direct candidates for the compliance investigation in RQ1. Within the dataset, the 749 rejection events and the extreme maximum throughput of 324 days are the strongest signals of non-standard behaviour and are investigated as anomaly candidates in RQ2. In terms of processing speed, prepaid expenses sit between the two extremes observed in other datasets faster than international claims (86.1 days) because they do not require the full post-trip declaration chain, but slower than domestic claims (11.2 days) because they still require a permit link and pre-trip approval. This three-way speed comparison across domestic claims, prepaid expenses, and international claims provides supporting evidence for the process redesign proposed in RQ3.

2.5 Payment Request

The Payment Requests dataset records non-travel work-related purchases submitted for reimbursement at UHasselt University, covering items such as office equipment, project materials, and other operational costs. Unlike the other three datasets, Payment Requests has no mandatory link to a travel permit and operates as a fully independent reimbursement process. It is the second largest dataset by row count and covers the widest date range of any single table.

2.5.1 Dataset Snapshot

Attribute	Value
Total rows (events)	36,796
Total unique cases (permits)	6,886
Time period	2017-01-09 to 2019-08-08
Columns	16
Unique activities	19

2.5.2 Activity Analysis

The 19 unique activities cover the full payment request lifecycle from employee submission through administration, budget owner, and supervisor approval to final payment. The activities with the highest value count are shown below;

Activity	Count
Request For Payment SUBMITTED by EMPLOYEE	7,504
Request For Payment FINAL_APPROVED by SUPERVISOR	6,343
Payment Handled	6,307
Request Payment	6,303
Request For Payment APPROVED by ADMINISTRATION	5,489
Request For Payment APPROVED by BUDGET OWNER	2,014
Request For Payment REJECTED by EMPLOYEE	1,085

The Request For Payment SUBMITTED by EMPLOYEE (7,504) exceeds the total case count of 6,886 by 618 events, which confirms that a subset of employees submitted multiple payment requests per case, likely after initial rejection and resubmission. The near-identical counts for Payment Handled (6,307) and Request Payment (6,303) indicate a minimal dropout at the final payment stage, which is consistent with all other datasets. The drop from Request For Payment

SUBMITTED (7,504) to Request For Payment FINAL_APPROVED by SUPERVISOR (6,343) represents a 15.5% attrition rate, meaning approximately 1 in 6 submitted payment requests never reached final supervisor approval. This is the most significant throughput loss point in the process.

2.5.3 Processing Time

Processing time is measured from the first recorded event to the last recorded event per permit case.

Metric	**Value (≈days)**
Mean throughput	12
Median throughput	8
Minimum throughput	0
Maximum throughput	406
Standard deviation	17

The mean throughput of 11.7 days is comparable to National Claims (11.2 days); both processes have no permit requirement and move directly from submission to payment through a straightforward approval chain. The median of 8 days means the typical payment request resolves within one and a half working weeks. However, the mean (11.7 days) is 46% higher than the median (8 days) and the maximum of 406 days is extreme for a process that typically completes under two weeks. A payment request taking 406 days represents over 13 months of processing time and is the strongest single throughput anomaly for Payment Request.

2.5.4 Key Data Quality Findings

- **Request For Payment REJECTED by MISSING appears prominently in rejection events.** MISSING represents an unidentified factor in approval or rejection steps performed by a resource not registered in the system. It means the university cannot verify who rejected a payment request.

- **Maximum throughput of 406 days for a non-travel payment request.** Given that the median resolution time is 8 days, a case taking 406 days represents a process failure of extraordinary scale. This warrants individual investigation as part of the RQ2 anomaly detection analysis.

2.5.5 Initial Insights

Payment Requests provides the clearest non-travel baseline in the dataset as its 8-day median throughput and permit-free structure confirm the university's approval chain functions efficiently when permit authorisation is not involved. The 6.2% rejection rate is the highest across all five datasets, it is the primary anomaly signal for RQ2 and suggests that non-travel purchase submissions have a higher error or non-compliance rate than travel-related submissions. The maximum throughput of 406 days is investigated as a critical anomaly later in this report. Unlike the other four datasets, Payment Requests does not directly inform RQ1 or RQ3; as it has no permit linkage requirement and no multi-claim structure but, it contributes meaningfully to the overall anomaly detection across the full dataset.

3. ER Diagram

3.1 Relationship

The diagram below illustrates an ER diagram for the travel and expense management dataset. It distinguishes between various entities, such as prepaid travel costs, payment requests, and both national and international expense claims. This representation shows how an employee navigates different processes established by Hasselt University.

An employee can submit multiple permits and reimbursement documents over time. International travel requires a travel permit; thus, both international claims and prepaid travel costs are linked to the corresponding permit through a foreign key. This means that multiple declarations or prepaid records can be associated with the same permit. In contrast, domestic claims and payment requests are independent of permits and are only linked to the employee.

Each permit or declaration is further organized as a case that consists of several timestamped events (activities). Together, these activities form the event logs used for process mining.

3.2 Key Identification

In the provided event logs, each record contains a *Case ID* column. This attribute is present in all tables and serves as the primary case identifier when importing data into process-mining tools. Because *Case ID* is not a business attribute but appears to be generated by the extraction procedure, we treat it as a surrogate key that the information system assigns to link events belonging to the same case across tables.

- **Business keys vs. technical keys**
  At the same time, each log contains business-specific identifiers (for example, *travel permit numbers,* domestic or international declaration numbers, prepaid identifiers) that can serve as natural

primary keys within their respective entities. In many instances, the values of *Case ID* and the business identifier coincide (for example, *Case ID* and *travel permit numbe*r may have identical values), indicating that the surrogate case id was derived from the business key for those cases.

- **The *id* key**
  Another attribute that exists in all tables is the id key. The id field uniquely identifies each row within a table (that is, each event), and therefore functions as an event-level technical key. However, it is not used to link tables to each other; inter-table relationships are modeled via *Case ID* and the business identifiers such as *travel permit number*, rather than via *id.*

3.3 Relationships, Keys & Entities

I. International Expense Claim

This entity represents reimbursement requests related to international travel.

Attributes:

- **CaseID/caseid (PK)**
- ***Permit ID (FK)*** referencing the travel permit entity.

II. National Expense Claim

This represents reimbursement requests for domestic travel expenses.

Attributes:

- **CaseID/caseid (PK)**

III. Prepaid Expense Claim

This entity represents prepaid expense claims, which are expenses paid before the trip, such as airfare or conference registration fees.

Attributes:

- **CaseID/Rfp_id (PK)**
- ***PermitId (FK)*** *referencing the travel permit that authorizes the trip*

IV. Payment Request Claim

This entity represents payment requests associated with expenses.

Attributes:

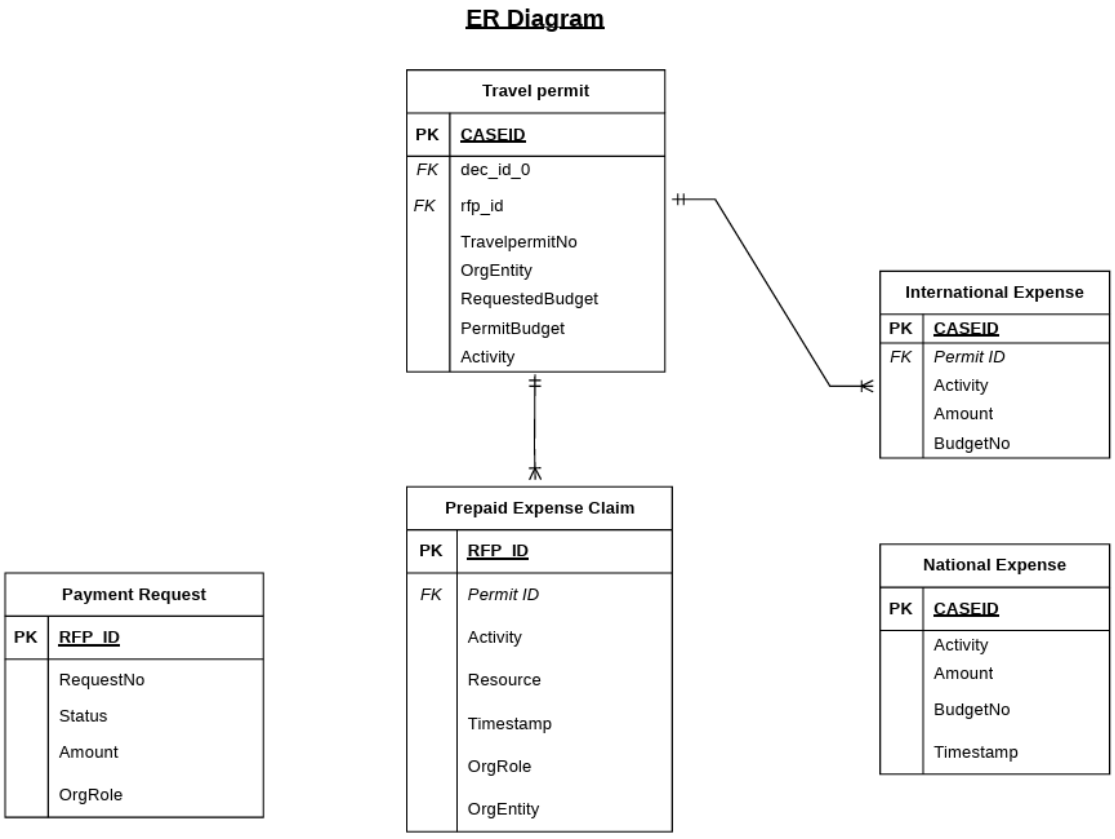
- **CaseID/Rfp_id(PK)**

IV. Travel permit

This represents travel permits associated with the international travel claim and pre-paid expenses

- **Case id/id as the PK**
- ***(dec_id_0, Rfp_id)* as the FK**

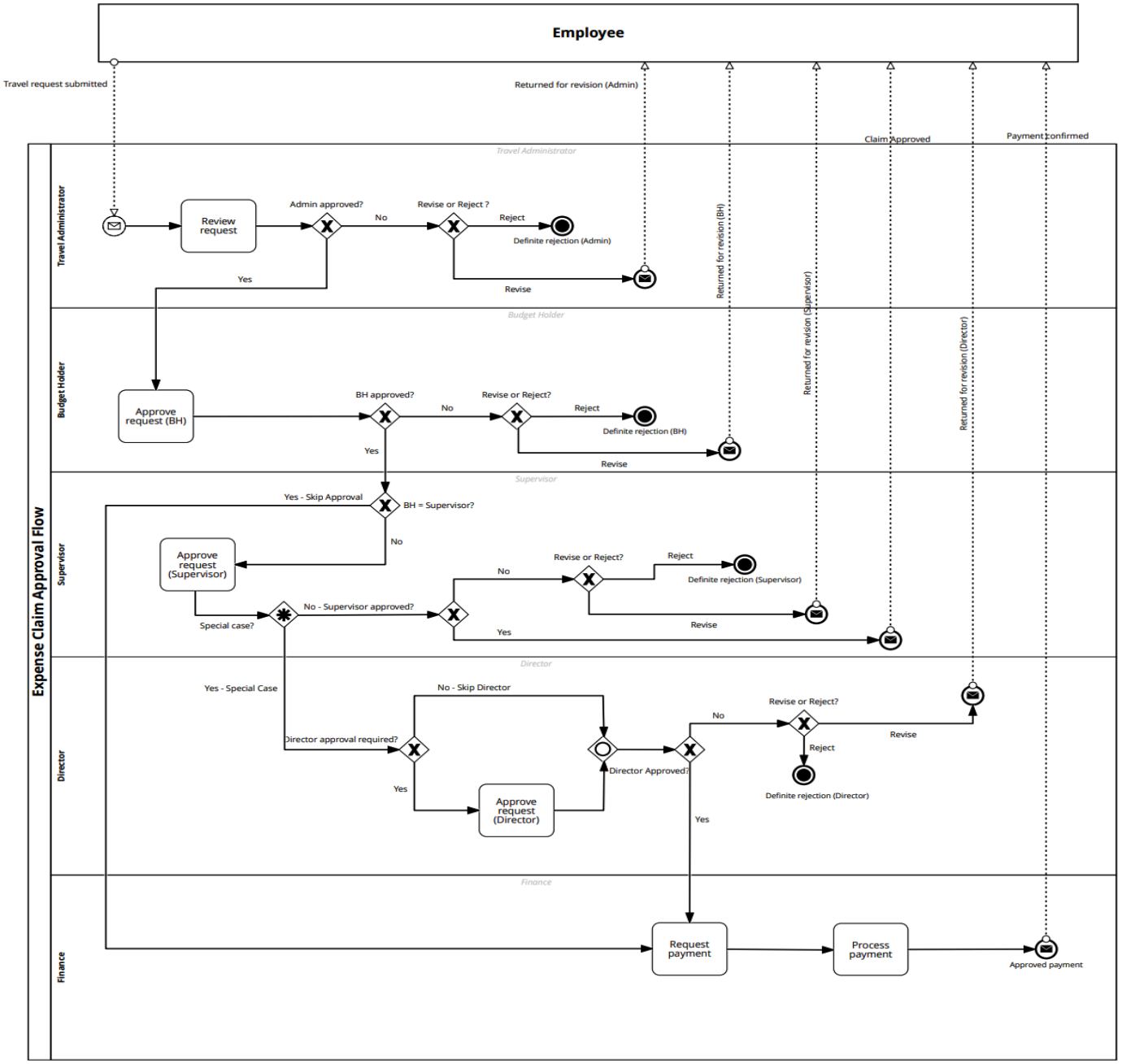
Note: ***Rfp_id*** and **CaseID/caseid (PK)** are the same key, withdifferent name



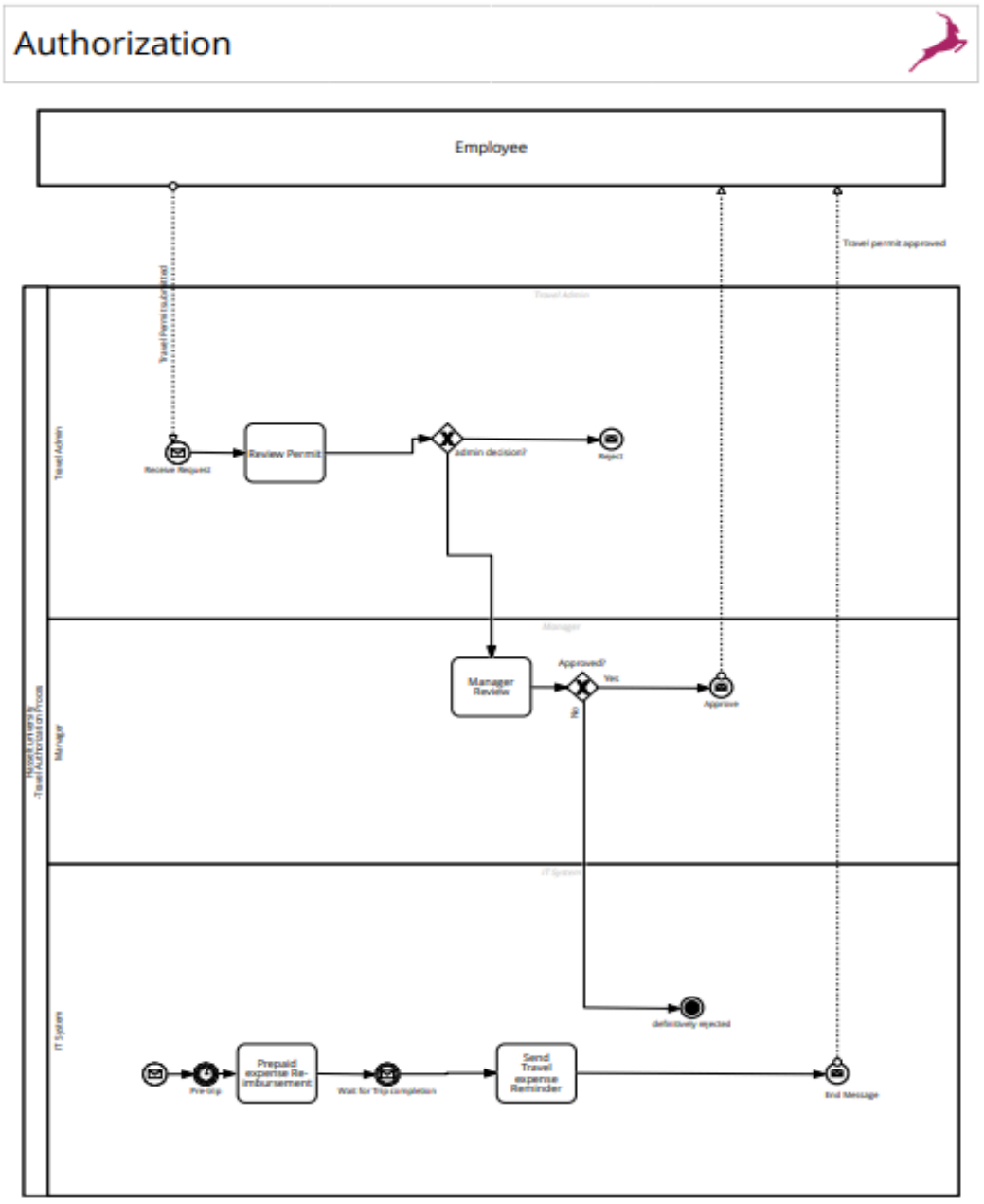
4. Business Process Modeling (BPMN)

In this section, the business process models are reconstructed based on the business understanding phase. These models will follow continuous refinement and validation during the data understanding and preparation phases using actual event log data.

4.1 Expense Claim Approval Flow



4.2 Travel Authorization Process



5. Findings and Analysis

This section presents the initial insights and analytical results for the research questions. Section 5.1 covers process analytics and compliance (RQ1) and section 5.2 covers data mining and anomaly detection (RQ2).

5.1 Process Analytics: Compliance Analysis

5.1.1 To what extent are expense claims correctly preceded by an approved travel authorisation?

The analysis used the International_expense_claims.csv dataset, which contains 72,151 event rows spanning 6,449 unique cases. The dataset records the full lifecycle of each international expense claim, including permit submission and approval activities, declaration events, trip activities, and payment processing.

To measure compliance, the analysis was structured into two branches that were subsequently joined on the common permit identifier.

Branch 1 extracted rows where the activity was either Permit FINAL_APPROVED by SUPERVISOR or Permit FINAL_APPROVED by DIRECTOR, using OR logic, and aggregated these per case using the minimum timestamp. This produced 6,001 cases where a valid final permit approval was recorded.

Branch 2 extracted rows matching Declaration SUBMITTED by EMPLOYEE and aggregated these per case using the minimum timestamp, producing 6,375 cases with a declaration event.

The two branches were joined using a left outer join, keeping all 6,375 declaration cases as the base. Each case was then classified as follows:

- No permit approval timestamp found: labeled "No permit approval."
- Permit approved before declaration submission: labeled "Compliant."

Cases without a declaration event (74 out of 6,449) were excluded from the compliance analysis, as they do not represent completed expense claims.

The results show that 5,937 cases (93.1%) were compliant, meaning the travel permit was approved before the expense claim was submitted. The remaining 438 cases (6.9%) had no final permit approval on record at all. No timing-violation cases were identified, meaning every case with a permit approval was recorded before the declaration was submitted.

Compliance Category	Case Count	Percentage of Total
Compliant	5,937	93.1%
No permit approval	438	6.9%
Total	**6,375**	**100%**

Among compliant cases, the majority of approvals were granted by the Supervisor. However, 640 cases required Director-level approval, reflecting the application of secondary approval rules for claims that exceeded budget thresholds or fell under specific organizational categories.

While a 93.1% compliance rate may appear strong at first glance, the 438 non-compliant cases represent a complete absence of the required control rather than a minor procedural delay. These cases are examined in detail in the following section.

5.1.2 Are there cases where expense claims exist without a corresponding authorization?

This question requires a distinction between two different types of compliance: structural compliance at the system level and substantive compliance at the process level. Understanding this distinction is what makes the findings significant.

At the structural level, all 6,449 cases in the dataset contain a populated travel permit number. The system enforces this as a mandatory field, so no claim can be submitted without entering a permit reference. In this sense, no claim is completely without a permit number.

At the process level, however, 438 cases (6.9%) were processed without the referenced permit ever reaching FINAL_APPROVED status. The permit number field was filled in, but the authorization it's supposed to represent never actually

occurred. The system verified that a value was entered, not that the value corresponded to a genuinely approved permit.

The Dummy Permit Pattern

Closer inspection of the 438 non-compliant cases reveals that they are not isolated oversights. All 438 cases share the same Trip_Time value of 2018-04-10T00:00 and are all linked to the same specific travel permit: permit 424. The probability of 438 separate cases all referencing the same permit and the same trip date by coincidence is negligible. This strongly suggests that permit 424, combined with a fixed dummy trip date, was used repeatedly as a workaround to satisfy the system's structural permit field requirement without going through the genuine authorisation process.

This finding reframes the interpretation of the 438 cases entirely. Rather than representing individual employee errors or oversights, they point to a systematic workaround that was discovered and repeatedly applied. Users found that by entering permit 424 and the corresponding dummy Trip_Time, they could progress through the process without obtaining a real FINAL_APPROVED permit.

Downstream Consequences

The payment workflow did not detect or block this workaround. Of the 438 non-compliant cases, 426 (97.3%) proceeded all the way to Payment Handled, meaning they were reimbursed despite the complete absence of a valid authorisation. Only 12 cases (2.7%) were stopped before payment was processed.

Outcome	Cases	% of 438
Payment Handled (paid out)	426	97.3%
No payment processed	12	2.7%
Total non-compliant cases	**438**	**100%**

This reveals a second control failure beyond the permit stage itself. The payment workflow operates independently of the permit workflow, with no integration point that checks permit approval status before releasing funds. The result is

that claims can enter and exit the process, resulting in financial reimbursement, without ever going through the authorisation step that the policy requires.

Financial Exposure

The 438 cases without permit approval represent a total of EUR 388,493.89 in submitted claims, of which approximately EUR 377,549 was actually paid out across the 426 cases that reached Payment Handled. The mean claim amount for non-compliant cases is 15.6% higher than for compliant cases, indicating that the workaround was not limited to small or low-value claims.

Compliance	Total Amount	Mean per Case
Compliant	EUR 4,557,123.69	EUR 767.58
No permit approval	EUR 388,493.89	**EUR 886.97**

Processing Time: Blending In With Compliant Cases

To assess whether non-compliant cases were processed differently in terms of speed, throughput time was measured from declaration submission to the final recorded event. The results show that the median throughput time is the same for both groups at nine to ten days, and the means are also very close.

Compliance	Mean Process Days	Median Process Days
Compliant	11.82 days	9 days
No permit approval	11.27 days	9 days

This finding is notable because it means the dummy permit cases do not stand out from a timing perspective. They move through the process at the same pace as properly authorised claims. This makes the workaround particularly difficult to detect through routine process monitoring, since cases using permit 424 produce throughput times that are entirely consistent with normal behaviour. The risk is therefore hidden behind otherwise normal-looking process performance.

The UNKNOWN Permit Risk

A separate but related issue was identified in the permit approval data. Among all travel permits that were finally authorised by a supervisor, 23 permits carry the value UNKNOWN as the permit identifier. Of these, 20 are linked to compliant declarations, meaning that from a timing perspective, the authorisation did precede the claim submission. However, 15 of those 20 cases progressed to Payment Handled, with an average claim amount of approximately EUR 4,305 per case, which is notably higher than the average for either the compliant or non-compliant groups.

These UNKNOWN permit cases represent a different category of risk. They are formally compliant in the sense that a permit approval is recorded before the declaration, but the absence of a traceable permit identifier means it is not possible to verify who approved what or to link the reimbursement back to a specific authorised trip. This undermines the auditability of those cases and creates a documentation gap even where the procedural sequence was technically followed.

Rework and Administrative Burden

The non-standard patterns described above also have a practical effect on workload. Across the full dataset, at least 1,755 cases include at least one rejection event and require resubmission, indicating that rework is a recurring rather than exceptional part of the process.

Rejected by	Compliant Cases	Non-Compliant Cases
Administrator	1,559	73
Supervisor	207	11
Total rejections	**1,766+**	**84**

Within the 438 non-compliant cases, rejections fall primarily on the administrative side, with 73 rejections by the administrator and 11 by the supervisor. This suggests that even when the dummy permit workaround is used, some cases still surface as problematic and require manual intervention to resolve. The concentration of administrative rejections in the non-compliant

group reinforces the idea that the workaround does not fully avoid scrutiny and instead shifts effort and decision-making onto administrators, who then have to manage the exceptions.

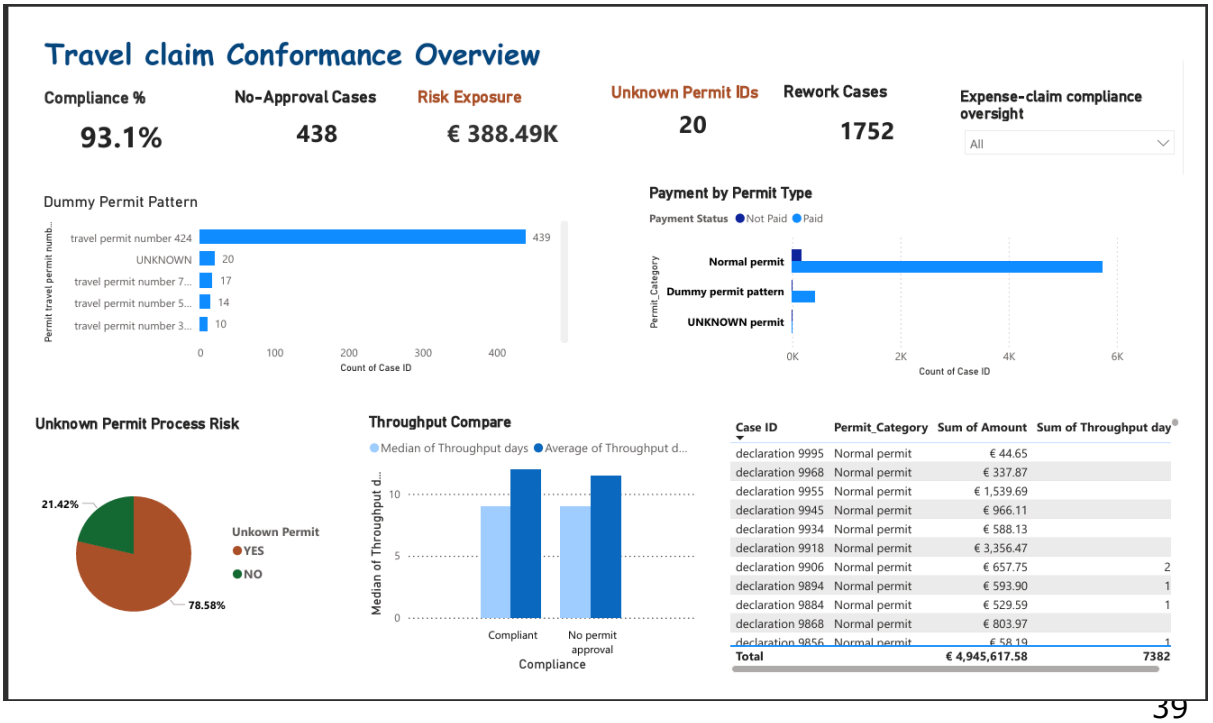
Amount Outlier Analysis

To further investigate the risk profile of non-compliant cases, IQR-based outlier detection was applied to the claim amount column. Thresholds were calculated separately for each compliance group so that each group was assessed against its own distribution rather than a single combined baseline.

Compliance	Outliers	Total Cases	Outlier Rate
Compliant	316	5,937	5.3%
No permit approval	34	438	**7.8%**
Total	**350**	**6,375**	**5.5%**

The outlier rate for non-compliant cases (7.8%) is 47% higher than for compliant cases (5.3%). Non-compliant cases make up 6.9% of all cases but account for 9.7% of all outliers. This confirms that the elevated mean claim amount seen in Table 3 is not simply a uniform shift across the group but is partly driven by a higher concentration of genuinely extreme values. Cases that bypassed the authorisation process were therefore not only more likely to be paid without oversight, but also more likely to involve unusually large amounts.

5.1.3 Dashboard Version 1



5.2 Data Mining: Anomaly Detection

Research question: Can anomalies or outliers be detected that may indicate non-standard or risky behavior?

5.2.1 Analytical Approach and Scope

RQ2 asks whether anomalies or outliers can be detected that indicate non-standard or risky behaviour. To answer this systematically, the analysis was structured around six specific hypotheses. Each hypothesis represents a different type of suspicious behaviour that could appear in the travel and expense data. Rather than searching the data randomly for anything unusual, each hypothesis gave the analysis a clear and testable focus.

The six hypotheses are:

1. **A1 (Unusual amounts):** Some expense claims may have amounts that are extremely high or extremely low compared to the rest. Very high amounts could indicate over-claiming or inflated submissions. Zero amounts suggest the claim was submitted without a recorded cost, a data quality problem.
2. **A2 (Overspending per permit):** When the total amount declared across all claims linked to a single travel permit exceeds the budget that was originally approved for that permit, the employee has spent more than was authorised. This is a direct financial control failure.
3. **B1 (Submission timing):** University rules require that expense claims are submitted within two months of the trip and only after the travel permit has been approved. Claims submitted too late or before the permit was formally approved violate these rules.
4. **B4 (Long throughput time):** Cases that take far longer to process than normal may indicate stalled approvals, contested claims, or cases that have been effectively abandoned in the system without a formal outcome.
5. **B5 (Repeated rejections):** Cases that are rejected and resubmitted multiple times place an extra burden on approvers and suggest that either the submission was non-compliant from the start or the approval criteria were unclear or inconsistently applied.
6. **C2 (Cost type inconsistency):** If the amount claimed is disproportionately higher relative to the type of cost being declared, it

may indicate miscategorization or inflated claims. This hypothesis could not be tested because the Cost Type column in the data was entirely unpopulated.

The analysis covers all five datasets with different levels of depth depending on each dataset's contribution to anomaly detection.

Dataset	**Role**	**Cases analysed**
International Expense Claims	Primary; all six hypotheses tested, combined scoring model applied	6,449
Travel Permits	Secondary; joined for overspending (A2), permit linkage (B1), rejection analysis (B5)	7,065
National Expense Claims	Supporting; hypothesis B5 (repeated rejections)	10,500
Travel Expenses (Prepaid)	Supporting; hypothesis A1 (unusual amounts), C2 attempted	2,099
Payment Requests	Supporting; hypothesis A1 (unusual amounts)	6,886

All hypothesis testing and anomaly detection was implemented in KNIME using Rule Engine nodes for flag creation, Numeric Outliers nodes for IQR-based detection, and GroupBy nodes for aggregation.

The compliance analysis in RQ1 identified 438 cases processed against a single dummy permit with Trip_Time 2018-04-10, which represents a systemic control bypass rather than an individual non-compliance. This finding contextualises the 89 cases lacking verifiable authorisation identified in RQ2. These cases comprise

of 23 cases with no permit reference and 66 cases where the permit number could not be matched to any valid record as both point to the same underlying weakness; the permit number field functions as a structural form requirement rather than a substantive financial control. The university's approval system can be satisfied by referencing a permit number without that permit representing a genuine approved authorisation.

These findings establish that the anomalies detected in RQ2 operate within a process context already compromised at the authorisation level, making the following hypothesis analysis not merely a statistical exercise but a direct investigation into the operational consequences of that control weakness.

5.2.2 Hypothesis Analysis

A1: Claims with unusually high or low amounts

- 19 international expense claims (0.29% of 6,449) exceeded €5,000, with the highest at €10,765.
- 274 claims (4.25%) were submitted with a zero amount flagged as data quality concerns.
- Payment Requests: one case reached €982,873 against the mean of €353; this is 2,785 times the average and the single most extreme financial outlier in the entire project.
- Travel Expenses: one case reached €507,168 against a mean of €1,523.
- Detection method: Rule Engine flag for amounts above €5,000 and Numeric Outliers node using IQR method. A case is classified as a high-value outlier when the claim amount exceeds €5,000, representing approximately six times the dataset mean of €806, a threshold selected to identify amounts that cannot be explained by typical international travel costs.

The €982,873 payment request represents an extreme financial risk that warrants individual investigation. At 2,785 times the dataset mean, this case cannot be explained by normal expenditure variation. The 274 zero-amount claims in the international dataset represent a data quality gap, claims were submitted and processed without a recorded monetary value, making it impossible to verify whether reimbursement was appropriate. From a financial control perspective, the 19 high-value claims represent the university's greatest

individual reimbursement risk, each requires verification that the amount corresponds to genuine, pre-authorised expenditure. The 274 zero-amount claims indicate a process gap where the system allowed claims to be submitted, reviewed, and in many cases paid without a monetary value ever being recorded.

A2: Overspending Per Permit

- 2,028 international expense claims (31.4% of 6,449) were linked to permits where the declared amount exceeded the approved budget.
- From the permits side, 1,894 of 7,065 permits (26.8%) were flagged as overspent.
- Detection method: Travel Permits and International Claims joined on permit number; Rule Engine flag applied to the overspent column from the permits dataset.

Nearly one in three international expense claims is linked to a permit where the declared amount exceeds the approved budget. This 31.4% overspending rate is not an isolated anomaly but rather, a systemic pattern indicating a structural weakness in the university's budget control process. Three possible explanations exist: budget approval limits are systematically set too low at the permit stage, employees are incurring costs beyond what was authorised without consequence, or the approval workflow does not enforce budget limits at the payment stage. From a process perspective, this finding means that the permit approval step which is intended to be the primary financial control for international travel is failing to contain expenditure in nearly one third of cases. Therefore, a review of how permit budgets are estimated and enforced is warranted.

B1: Claim Submission Timing

This section examines two related compliance dimensions: a. whether a valid permit existed before the claim was submitted, and b. whether the claim was submitted within the required timeframe after permit approval.

a. Permit Existence Check

Permit authorisation analysis identified two distinct categories of non-standard cases. The first category comprises 23 cases (0.36%) where the travel permit number field was recorded as UNKNOWN, indicating that no permit was

referenced at the point of claim submission. These represent the most obvious authorisation failures claims submitted with no attempt to link an approved permit. The second category comprises 66 cases (1.02%) where a permit number was recorded but did not match any valid permit record in the travel permits dataset. Together these 89 cases (1.38% of 6,449 international claims) represent the full population lacking confirmed prior authorisation, connecting directly to the 426 reimbursed-without-approval cases identified in RQ1.

b. Submission Timing Analysis

- No direct violations were found where a claim was submitted before permit approval within the 5,937 testable cases.
- 2,037 cases (34.3% of testable cases) were submitted more than 60 days after permit approval, violating the university's two-month submission deadline.
- 108 cases (1.8%) were submitted on the exact same day as permit approval, same-day submissions that may indicate rushed or retrospective authorization.
- 512 cases could not be tested due to unavailable permit approval timestamps.
- Detection method: Submission and approval timestamps extracted from the event log, joined on Case ID, date difference calculated using the Date and Time Difference node.

While no employee submitted a claim before their permit was approved within the testable population, the analysis reveals three distinct compliance concerns. First, 89 cases lacked any verifiable prior authorisation. Second, over one third of testable cases (34.3%) violated the two-month submission deadline, suggesting the rule is either not communicated effectively, not enforced by the system, or routinely disregarded by the organisation. Third, the 108 same-day submissions indicate possible retrospective authorisation where the permit was approved simultaneously with or after the expense was incurred. From a process perspective, these findings reveal that the prior authorisation requirement has three exploitable weaknesses; it can be bypassed through unmatched permit references, it does not enforce timely submission after travel, and same-day approvals may allow claims to be authorised after the fact rather than in advance.

B4: Throughput Time Outliers

- 404 international expense claims (6.3%) were identified as throughput outliers.
- Overall distribution: minimum 6 days, maximum 742 days, mean 86.1 days, median 66 days.
- Across all datasets: Travel Permits maximum 1,190 days (over three years for a single authorisation).
- Detection method: GroupBy on Case ID for first and last event timestamps, Date and Time Difference node for throughput calculation, Numeric Outliers node with IQR multiplier 1.5 and full data estimate.

A permit taking 1,190 days and an international claim taking 742 days represent process failures of extraordinary scale. These cases were not simply slow, they were effectively abandoned in the system without a resolution. For the employees involved, this means waiting over two years for a reimbursement outcome, creating financial uncertainty and eroding trust in the administrative process. For the university, unresolved cases represent open financial liabilities that cannot be properly accounted for until they are closed. The 404 throughput outlier cases (6.3%) represent a meaningful population where the approval process broke down. Cases with repeated rejections identified under B5 are the most likely drivers of extended throughput, as each rejection cycle adds days or weeks to the total processing time.

B5: Repeated Rejections

- 434 international claims (6.7%) were flagged as high activity cases with 15 or more process events, indicating repeated rejection and resubmission cycles.
- 1,755 international claims (27.2%) experienced at least one rejection.
- 1,912 travel permits (27.1%) experienced at least one rejection.
- 1,301 national claims (12.4%) experienced at least one rejection — counterintuitively the highest single rejection count was 12 on one case in this dataset.
- Detection method: Event log filtered for rows containing REJECTED, grouped by Case ID with Count aggregation.

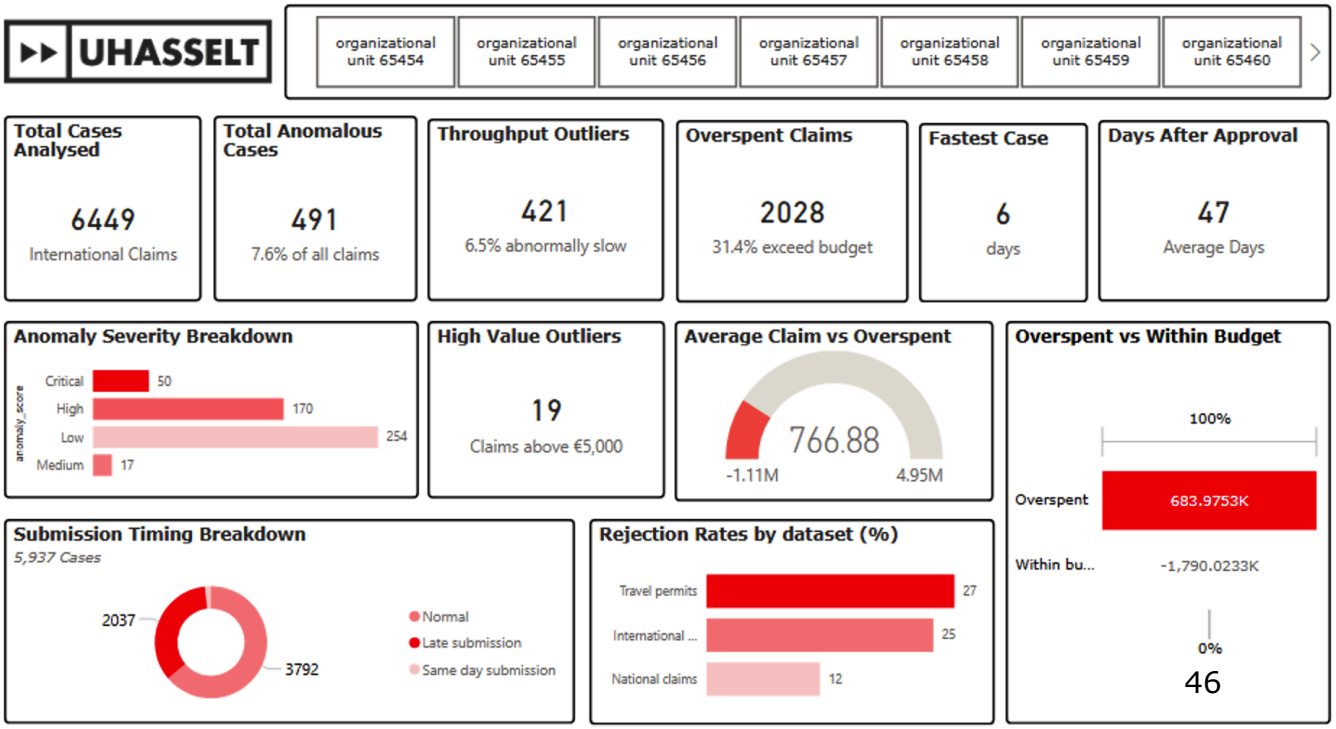
Over one quarter of all international claims and travel permits experienced at least one rejection. Each rejection cycle creates direct and measurable rework costs. The same approvers must review the same case multiple times, employees must revise and resubmit their claims, and the overall process extends significantly, directly contributing to the throughput outliers identified under B4. The counterintuitive finding that national claims have a lower overall rejection rate (12.4%) but a higher single-case maximum (12 rejections) suggests a different pattern in domestic claims rather than widespread low-level rejection. A small number of domestic cases are persistently problematic, possibly reflecting individual employees or departments with recurring compliance issues. The university should investigate whether rejection patterns concentrate in specific departments or among specific approvers, as this would indicate a training or communication gap rather than a systemic process failure.

C2: Cost Type Inconsistency

- The Cost Type column in the Travel Expenses dataset contained all zero values; the field was never populated in the data.
- The hypothesis could not be tested as designed.

The absence of cost type data represents a systemic data quality gap. Without cost category information it is impossible to assess whether expense amounts are proportionate to the type of cost being claimed. This is a missing data field that, if populated, would significantly strengthen future anomaly detection capability.

5.2.3 Dashboard Version 1.0



5.3 Process Redesign: Throughput Optimization

Research question: How can the end-to-end throughput time be improved when multiple expense claims are linked to a single travel authorization? How would this redesigned process be represented in BPMN?

5.3.1 Motivation for Redesign

The compliance and anomaly analyses conducted in RQ1 and RQ2 identified three major interconnected process weaknesses that the current workflow fails to address. First, RQ1 identified 438 cases processed against a single dummy permit (permit 424), all sharing the trip date 2018-04-10, with 426 subsequently paid out without valid authorisation, representing €377,549 in uncontrolled reimbursements. RQ2 independently identified 89 cases lacking verifiable authorisation through permit linkage analysis, confirming the same underlying control weakness from a different analytical angle. Together these findings demonstrate that independently processed claims are never cross-checked against a single verified trip-level authorisation. Furthermore, the B4 throughput analysis found a mean processing time of 86 days per claim with 404 cases flagged as outliers and a maximum of 742 days, while B5 identified that 27.2% of international claims experienced at least one rejection, each adding a complete approval cycle to an already lengthy process. Lastly, the A2 analysis found that 31.4% of claims were linked to overspent permits, indicating that approvers authorising individual claims have no visibility of cumulative trip expenditure at the point of each decision. These weaknesses share a common structural cause the current process treats each expense claim as an independent process instance, with no coordination at the trip level.

5.3.2 Current Process (As-Is)

Under the current process, each expense claim linked to a travel permit is processed independently through the full approval workflow; administration review, budget holder approval, supervisor approval, director approval for special cases, and then, payment. When a single trip generates multiple claims, each triggers a separate approval cycle involving the same approvers reviewing costs from the same trip on separate occasions. This structure means a single invalid permit reference can pass through multiple independent approval cycles

unchallenged as demonstrated by permit 424. This passed through 438 separate approval cycles and resulted in €377,549 in payments before the workaround was identified through the analysis rather than through the process itself. The total trip costs remain invisible to any individual approver until all claims are paid, and each rejection cycle extends the overall process independently rather than being resolved once at the trip level.

5.3.3 Redesign Proposal (To-Be)

The proposed redesign introduces a claim bundling mechanism at the permit level. Following permit approval, employees submit all expense claims within a defined waiting window after the trip end date. Once the window closes or the employee signals that all claims have been submitted, the bundle enters the approval workflow as a single coordinated unit. One administration review, one budget holder approval, and one supervisor sign-off cover all claims simultaneously, allowing the approver to assess total trip expenditure against the approved permit budget in a single decision. Payment is then issued as one consolidated instruction covering all approved items in the bundle.

This redesign addresses all three weaknesses identified in section 5.3.1. The trip-level review replaces independent claim-by-claim processing, eliminating the compliance gap that allowed dummy permits to pass through multiple approval cycles unchallenged. Under the redesigned process, the approver would verify permit validity once against the entire bundle before any payment is authorised; a control point that does not exist in the current workflow, where the payment system operates independently of permit approval status. Cumulative trip costs are visible to the approver at the point of approval, directly addressing the overspending pattern identified under A2. And the consolidation of approval steps into a single cycle reduces the number of independent rejection cycles, directly targeting the throughput inefficiency identified under B4 and B5.

However, this would systematically lead to some trade-offs and limitations. The waiting window introduces a fixed delay for employees with only one claim, they must wait for the window to close before processing begins. To mitigate this, an early trigger should be available; if the employee signals bundle completion before the window expires, processing begins immediately. Additionally,

employees may discover a missing receipt after the bundle has entered the approval workflow. The redesigned process must therefore include a resubmission mechanism modelled as an interrupting boundary event that allows late additions without restarting the entire approval cycle. Finally, the 2017 trial phase data cannot be used to measure baseline throughput for comparison purposes, as the limited two-department scope makes year-on-year comparison unreliable.

5.3.4 Redesigned BPMN

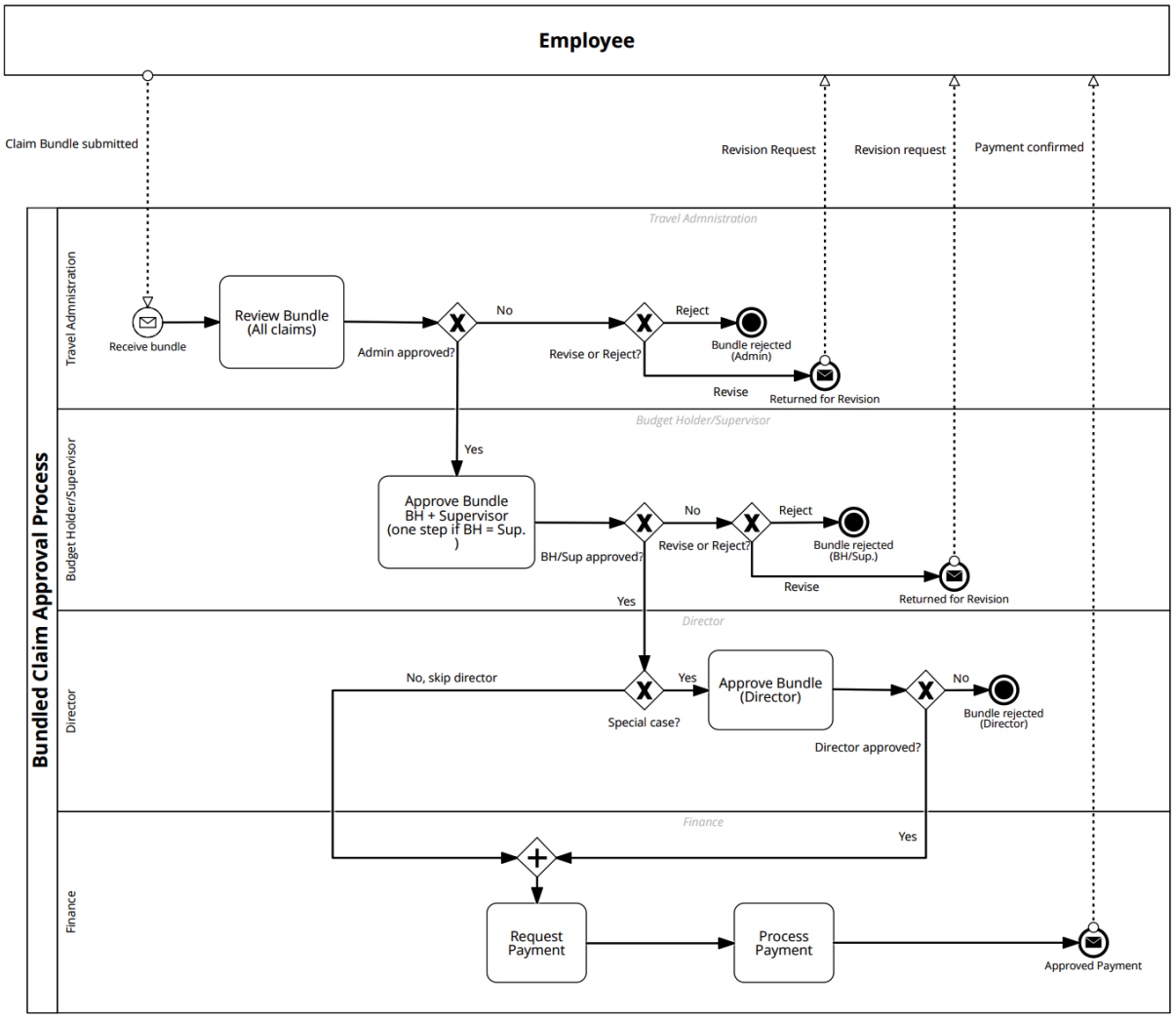


Figure 5.3.4

6. Discussion, Limitations and Recommendations

The synthesises of the findings presented in Chapter 5 are discussed in this chapter. This discussion draws connections across the three research questions, examines what the combined evidence reveals about the university's travel and expense management system as a whole, and identifies the practical implications for process improvement. The chapter also acknowledges the limitations of the analysis and concludes with concrete recommendations for the university.

6.1 Synthesis of Findings

The three research questions collectively reveal a travel and expense management system that is structurally compromised at every stage of its control chain. RQ1 established that the permit authorisation requirement, the system's primary financial safeguard can be bypassed entirely through a dummy permit reference, with 438 cases processed against permit 424 and €377,549 paid out without valid authorisation. RQ2 confirmed this control weakness from a different angle, identifying 89 cases lacking verifiable authorisation through independent permit linkage analysis, a 31.4% overspending rate indicating that budget limits are not enforced at the claim level, and a 34.3% late submission rate showing that procedural deadlines are routinely disregarded. RQ3 demonstrated that these failures share a common structural cause independent of the claim processing and proposed a bundling mechanism that addresses compliance oversight, budget visibility, and throughput efficiency simultaneously.

The combined picture is not one of isolated employee errors but of a system that structurally enables non-compliance. The dummy permit workaround was used 438 times and moved through the process at the same speed as legitimate claims with a median of nine days making it undetectable through routine performance monitoring. This is the most significant finding across the entire analysis; the system's internal controls cannot distinguish compliant from non-compliant cases based on process behaviour alone.

6.2 Business Problem Analysis

BP1: Data Quality and 2017 Structural Break

The 2017 trial phase involved only two departments and predates multiple revisions to the process rules. This structural break has direct consequences for the analytical findings. Compliance rates, rejection frequencies, and throughput distributions in 2017 are not directly comparable to 2018 without accounting for the limited scope and different process rules in operation. Where anomalies were detected in the combined dataset, they cannot be attributed to behavioural non-compliance without first verifying whether the relevant business rules were in force at the time. The dummy permit pattern identified in RQ1 was specifically dated to 2018 (Trip_Time 2018-04-10), which strengthens its interpretation as a deliberate workaround rather than a trial-phase artefact. Future analyses should partition data by year as a primary control variable and apply process rules only to the period in which they were active. Compliance rates and anomaly counts reported in this analysis should be interpreted as 2018-centric findings given this limitation.

BP2: Independent Audit

The finding in RQ1 that 426 of 438 non-compliant cases were paid out without detection reveals a direct failure of the verification function. The payment workflow operates independently of the permit approval workflow with no integration point that checks permit validity before releasing funds. This means the audit department regardless of its organisational independence is reviewing cases after payment has already occurred rather than before. Post-payment auditing cannot prevent financial exposure; it can only detect it retrospectively.

In RQ2, the finding that non-compliant cases carry a 7.8% outlier rate compared to 5.3% for compliant cases confirms that the cases most likely to require scrutiny are also the ones most likely to involve larger amounts. For the audit department to fulfil its mandate of enforcing credibility standards, a pre-payment permit validity check must be embedded in the approval workflow rather than relying on post-process review. The redesign proposed in RQ3 provides this control point by requiring bundle-level permit verification before any payment instruction is issued.

BP3: Rework and Delays

The anomaly detection analysis directly quantifies the rework problem as over one quarter of international claims (27.2%) and travel permits (27.1%) experienced at least one rejection, with individual cases reaching 14 rejection cycles. The B4 throughput analysis found a mean processing time of 86 days and 404 cases exceeding the IQR-based outlier threshold, with one case unresolved for 742 days. The B1 analysis found that 34.3% of claims were submitted after the 60-day deadline, indicating that employees are not engaging with the process on the timeline it requires.

These findings collectively point to BP3,which seems to be an insufficient employee understanding of the process requirements, as a systemic rather than an individual problem. When over one third of employees miss the submission deadline and over one quarter generate at least one rejection, the cause is more likely to be inadequate communication of requirements or insufficient system enforcement than individual negligence. The process redesign partially addresses this by introducing a waiting window that structures the submission timeline, but it does not resolve the underlying communication gap. The university should complement the process redesign with targeted employee guidance on submission requirements, particularly for departments with disproportionately high rejection rates identified in the anomaly analysis.

BP4: Conflicting Roles

The business rules acknowledge that the budget holder and supervisor are sometimes the same person, collapsing two approval steps into one. The B5 analysis in anomaly detection found that 27.2% of international claims experienced at least one rejection, and the RQ1 analysis identified that even non-compliant cases generated administrative rejections. This suggests that the approval chain does not consistently apply the same standards across all cases. When the budget holder and supervisor roles are held by the same individual, the dual accountability function of the two-step approval is eliminated. One approver cannot provide independent financial oversight and operational sign-off simultaneously, creating a single point of failure in the control chain. This role conflict is most consequential in the context of the dummy permit findings; a budget holder who is also the supervisor has no independent counterpart to

challenge an unusual permit reference. The RQ3 redesign preserves the role-collapse rule for operational efficiency but does not resolve the accountability gap it creates. A formal policy requiring escalation to director level when budget holder and supervisor roles coincide on high-value bundles would mitigate this risk without removing the efficiency benefit.

6.3 Limitations

Four analytical and technical limitations constrain the conclusions drawn in this report. First, the 2017 structural break means findings cannot be reliably generalised across both years without year-level controls. Second, the dummy permit pattern was identified through process analysis but cannot be confirmed as deliberate fraud without an individual case investigation. Therefore, administrative error or system misconfiguration remain alternative explanations.

Additionally, 512 cases could not be tested for B1 timing compliance due to missing permit approval timestamps, representing a data completeness gap that may conceal additional violations. Furthermore, the Cost Type field was entirely unpopulated in the travel expenses dataset, preventing C2 hypothesis testing and limiting the scope of cost-category anomaly detection.

6.4 Recommendations

Four recommendations follow directly from the findings.

I. **Embed a pre-payment permit validity check**. The payment workflow must verify that the linked permit has reached FINAL_APPROVED status before releasing funds. This single integration point would have blocked 426 of the 438 dummy permit payments identified in RQ1.

II. **Enforce the 60-day submission deadline systematically**. Automated reminders at 30 and 45 days after permit approval, combined with a system-level block on submissions beyond 60 days, would address the 34.3% late submission rate identified in RQ2 without requiring manual enforcement.

III. **Implement the bundled claim approval process.** The RQ3 redesign should be implemented for all permits with more than one linked claim. The consolidation of approval cycles reduces rework, improves budget

visibility, and closes the compliance gap that allows independent claims to bypass trip-level authorisation checks.

IV. **Escalate to director level when roles conflict.** Cases where the budget holder and supervisor are the same individual and the bundle value exceeds a defined threshold should be automatically escalated to director approval, restoring independent financial oversight without disrupting the standard workflow for lower-value cases.

6.5 Conclusion

The university's travel and expense management system faces four interconnected failures: a permit authorisation control that can be bypassed without detection, a payment workflow that operates independently of that control, a submission culture where deadlines and requirements are routinely not met, and an approval chain with structural role conflicts that reduce independent oversight. These failures are not independent, they reinforce each other.

Moreover, the dummy permit workaround succeeded precisely because the payment system does not check permit validity, because independent approval was compromised by role overlap, and because the process produced no timing signal that would distinguish a fraudulent submission from a legitimate one.

To sum up, addressing any one of these failures in isolation will not resolve the systemic risk. The recommendations above are designed to be implemented together, with the bundled claim approval redesign as the structural foundation on which the other controls operate.